

S.No. : 258

NEC 4101

No. of Printed Pages : 07

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43401

Roll
No.

1	2	5	0	4	3	2	2	6	6
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B. Tech. Examination, 2025-26

(Odd Semester)

BASIC ELECTRONICS ENGINEERING

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION –A

1. Attempt all parts of the following : $8 \times 1 = 8$

- Give name of three pentavalent impurities.
- State mass action law.
- Define α and β . Also write down the relation between α and β .

[P. T. O.]

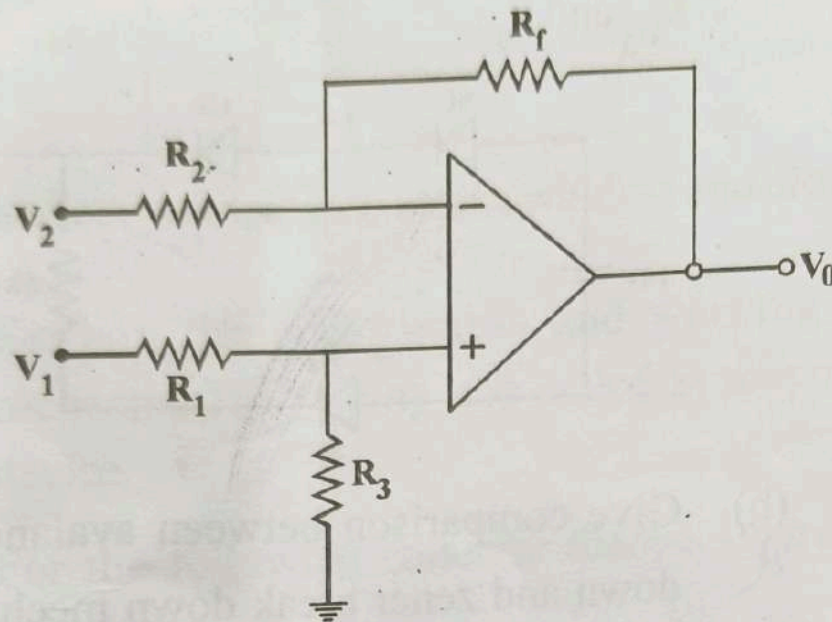
- (d) What do you mean by I_{CB_0} ?
- (e) Draw the circuit diagram of integrator using OP-amp.
- (f) What is the gain of voltage follower?
- (g) Find 2's complement of $(0000)_2$.
- (h) Find gray code of $(01010011)_2$.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Explain the effect on depletion width when P-N junction is :
 - (i) Forward biased
 - (ii) Reverse biased
- (b) Explain the working of NPN transistor. In n-p-n transistor with $\alpha = 0.97$ is operated in CB configuration. If emitter current is 3 mA and reverse saturation current is $10 \mu A$. Find out base and collector current.

(c) Derive the output for following circuit diagram:



(d) Perform the following conversion :

(i) $(123.45)_{10} = (\quad)_2$

(ii) $(1010011.011)_2 = (\quad)_{10}$

(iii) $(2DF.A9C)_{16} = (\quad)_8$

(iv) $(327.125)_8 = (\quad)_{16}$

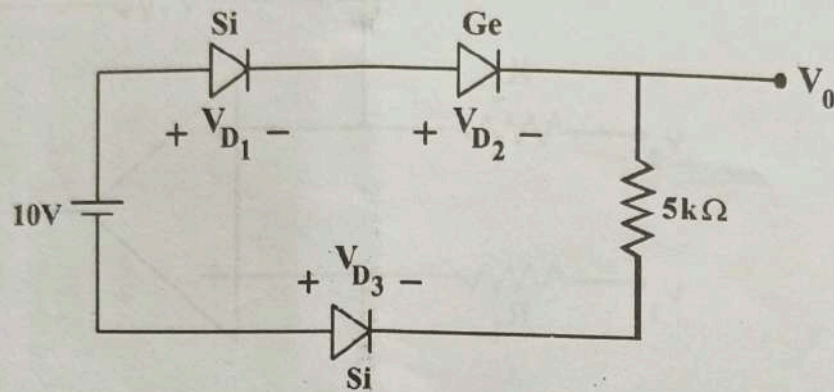
SECTION-C

Note :- Attempt all questions. Attempt any two parts from each questions.

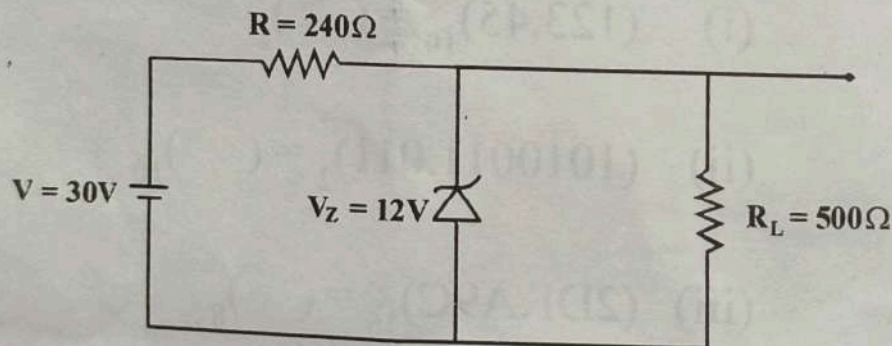
$$8 \times 5 = 40$$

[P. T. O.]

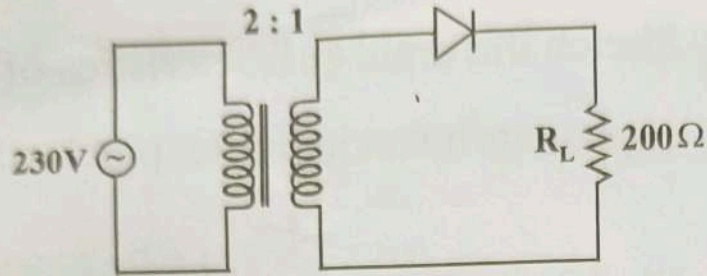
3. (a) For the following circuit diagram, find V_{D1} , V_{D2} , V_{D3} and V_0 :



- (b) Give comparison between avalanche breakdown and zener breakdown mechanism. For the following zener diode network calculate power dissipated by zener diode:



- (c) Explain working of half wave rectifier circuit with proper output wave form. A half wave rectifier circuit with a transformer coupled input is shown below. Determine the maximum and average values of power delivered to the load:



neglect voltage drop across winding and diode.

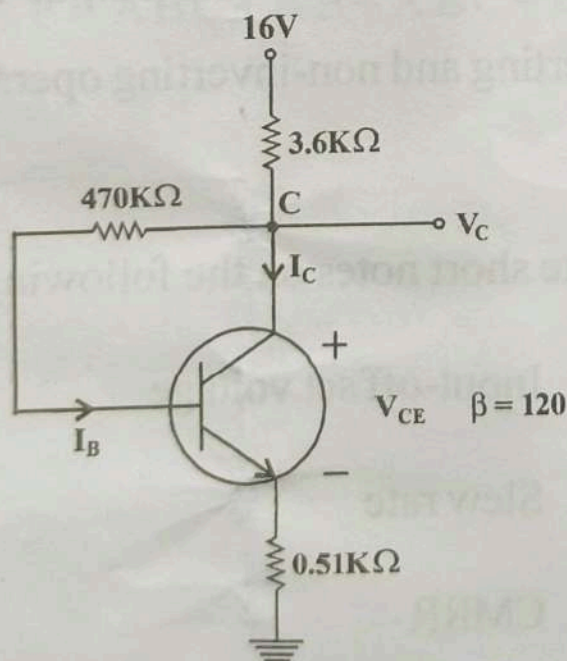
4. (a) Explain the construction and working of n-channel JFET. Why it is called as unipolar device.

(b) For the following collector-feedback circuit calculate :

(i) I_B

(ii) I_C

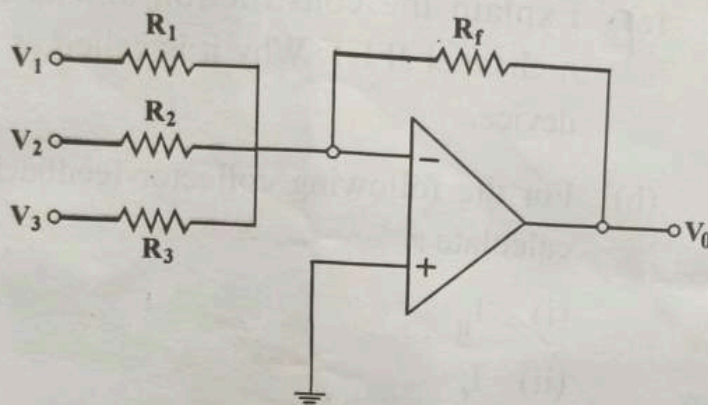
(iii) V_C



[P. T. O.]

- (c) Sketch the drain characteristics of a JFET and explain its behaviour after pinch-off.

5. (a) Find the output voltage (V_0) for the following OP-Amp circuit :



- (b) Find the expression for output voltage V_0 of inverting and non-inverting operational amplifier.

- (c) Write short notes on the following :

- (i) Input offset voltage
- (ii) Slew rate
- (iii) CMRR

6. (a) Perform the following operations and find the complement and

(45)

- (b) Explain De Morgan's theorem to simplify the expression :

$$\overline{A + BC}$$

- (c) Why NAND & NOR gates are called universal gates. Implement the function $F = ABC + \dots$ using only NAND gates.

$$F = ABC + \dots$$

6. (a) Perform the following subtraction using 1's complement and 2's complement :

$$(45)_{10} - (29)_{10}$$

- (b) Explain Demorgan's theorem. Use Demorgan's theorem to simplify following Boolean expression :

$$\overline{\overline{A + B\overline{C}} + \overline{D(E + \overline{F})}}$$

- (c) Why NAND & NOR gates are called as universal gates. Implement the following function using NAND gate only :

$$F = ABC + \overline{A}B + A\overline{B}\overline{C} + BC$$
